

## **Forced oscillations**

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This experiment was performed in collaboration with Joel Cunnah

### **Abstract**

The amplitude and phase of a vibrating mechanical system with a driving force applied to a phosphor-bronze bar were measured as a function of the driving frequency, for both damped and undamped oscillations. The decay of the system when undergoing free oscillations was observed, and the decay constant of the oscillations was measured. The decay was found to be exponential. The resonant frequency and quality factor of the system were calculated for different levels of damping. The quality factor and peak amplitude of the system were found to decrease with increased levels of damping as predicted by the theory of resonance.

## 1. Introduction

The oscillations in this experiment are an example of simple harmonic motion, this means the oscillations repeat with a certain period. The restoring force of the motion is directly proportional to the displacement and acts in the direction directly opposite the displacement [1]. The displacement can be found by solving the equation of motion. The system trades energy back and forth between the potential energy of the bar as it bends and the kinetic energy of the bar as it oscillates [4]. All vibrating systems have a natural frequency that they vibrate at if no external forces are present. If a driving force is applied with a frequency that is close to this natural frequency then the amplitude of the oscillations will increase dramatically as the force adds energy to the system, this phenomena is known as resonance [2].

## 2. Theory

### 2.1 Simple Harmonic Motion

Simple harmonic motion is described by a differential equation of the form

$$m \frac{d^2x}{dt^2} = -kx$$

where  $x$  is the displacement,  $t$  is the time,  $m$  is the mass and  $k$  is a constant [5].

The standard solution to this equation is

$$x(t) = A \sin(\omega_0 t) + B \cos(\omega_0 t)$$

where the natural frequency  $\omega_0 = \sqrt{k/m}$

### 2.2 Damped oscillator

For a damped oscillator the damping force is proportional to the velocity,  $F = -bv$ , where  $b$  is a constant and  $v$  is the velocity.

The equation of motion then becomes

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

the solution to this equation is

$$x(t) = \exp(-\gamma t/2) \sin(\omega t + \phi)$$

where  $\gamma = b/m$  and  $\omega^2 = \omega_0^2 - \gamma^2/4$ . This fails when  $\gamma$  reaches  $2\omega_0$ , where there is critical damping.

## 2.3 Driven Oscillator

For a driven oscillator the driving force was assumed to have the form  $F = F \sin(\Omega t)$ .

This leads to the equation of motion

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = F \sin(\Omega t)$$

This has two different parts to its solution, a transient described by the undriven damped oscillator

$$x(t) = \exp(-\gamma t/2) \sin(\omega t + \phi)$$

where  $\phi$  is the phase difference  
and a steady state solution

$$x(t) = A \cos(\Omega t - \phi)$$

Using complex exponentials we can derive an expression for the square of the amplitude, which is proportional to the power

$$A^2 = \frac{(F/m)^2}{(\Omega^2 - \omega_0^2)^2 + \gamma^2 \Omega^2}$$

This shows that the amplitude will increase at the natural frequency, meaning resonance will occur.

## 2.4 Quality factor

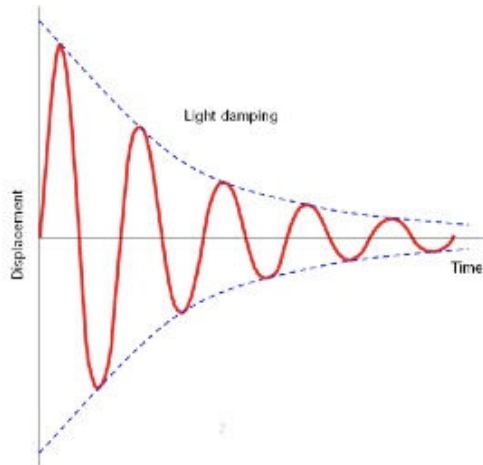
The behaviour of the denominator in this expression depends mainly on  $\omega_0$  and  $\gamma$ , the ratio of these two numbers give the quality factor  $Q = \omega_0/\gamma$  [Equation 1]. A vibrating system with a high quality factor resonates with a greater amplitude at its natural frequency than one with a low quality factor.

The quality factor is also given by  $Q = f^0/\Delta f$  [Equation 2], where  $f^0$  is the resonant frequency and  $\Delta f$  is the bandwidth, the range of frequencies for which the energy is at least half its peak value.

The quality factor gives some indication of how fast energy is lost from a vibrating system, its definition is  $2\pi$  times the ratio of the total energy stored divided by the energy lost in a single cycle.

## 2.5 Free oscillations

Free oscillations means there is no driving force, the system oscillates at its natural frequency. In theory the oscillations should remain at the same amplitude but in reality there is always some damping due to friction so the system behaves like a lightly damped oscillator.



The amplitude of the oscillations falls off exponentially,  
 $A = A_0 \exp(-\gamma t/2)$  [2]

This exponential decay can be linearised to the form  
 $y = mx + c$ , resulting in

$$\ln A = -\gamma t/2 + \ln A_0 \quad [\text{Equation 3}]$$

## 3. Method

In this experiment the oscillator is a phosphor-bronze bar which is rectangular in cross section. One end of the bar is clamped and the other has an aluminum plate attached to it. An elastic band wrapped round the bar is attached to a rod below it. This rod is driven by a loud speaker coil powered by a function generator. The frequency of the driving force can be varied using the function generator.

A coil of insulated copper wire is wrapped around the bar. As the bar moves the coil moves between the poles of a fixed horseshoe magnet, inducing a voltage through the coil due to Faraday's law. This voltage can be viewed on an oscilloscope and is directly proportional to the velocity of the oscillating bar.

When the bar oscillates the aluminum plate attached to one end moves between the poles of an electromagnet. When the electromagnet is turned on the changing magnetic field in the moving aluminium plate induces a circular current known as an eddy current. Lenz's law states that this eddy current creates a magnetic field that opposes the magnetic field that induced the current. Thus a damping force is created which is proportional to the velocity of the bar [3].

A check was made to ensure that when the bar oscillates at resonant frequency the amplitude of the oscillations is not so large that the aluminum plate collides with the bottom of the electromagnet.

## 4.Results

### 4.1 Velocity-resonance curve

The drive voltage and the amplified voltage from the coil were displayed as sinusoidal waves on the oscilloscope. The amplitude of the oscillation was measured in arbitrary units for a range of driving frequencies, between 10 and 20 Hz. The amplitude of the response signal is directly proportional to the velocity of the bar. A graph of amplitude against driving frequency was plotted. The uncertainty in amplitude was recorded and used to plot error bars.

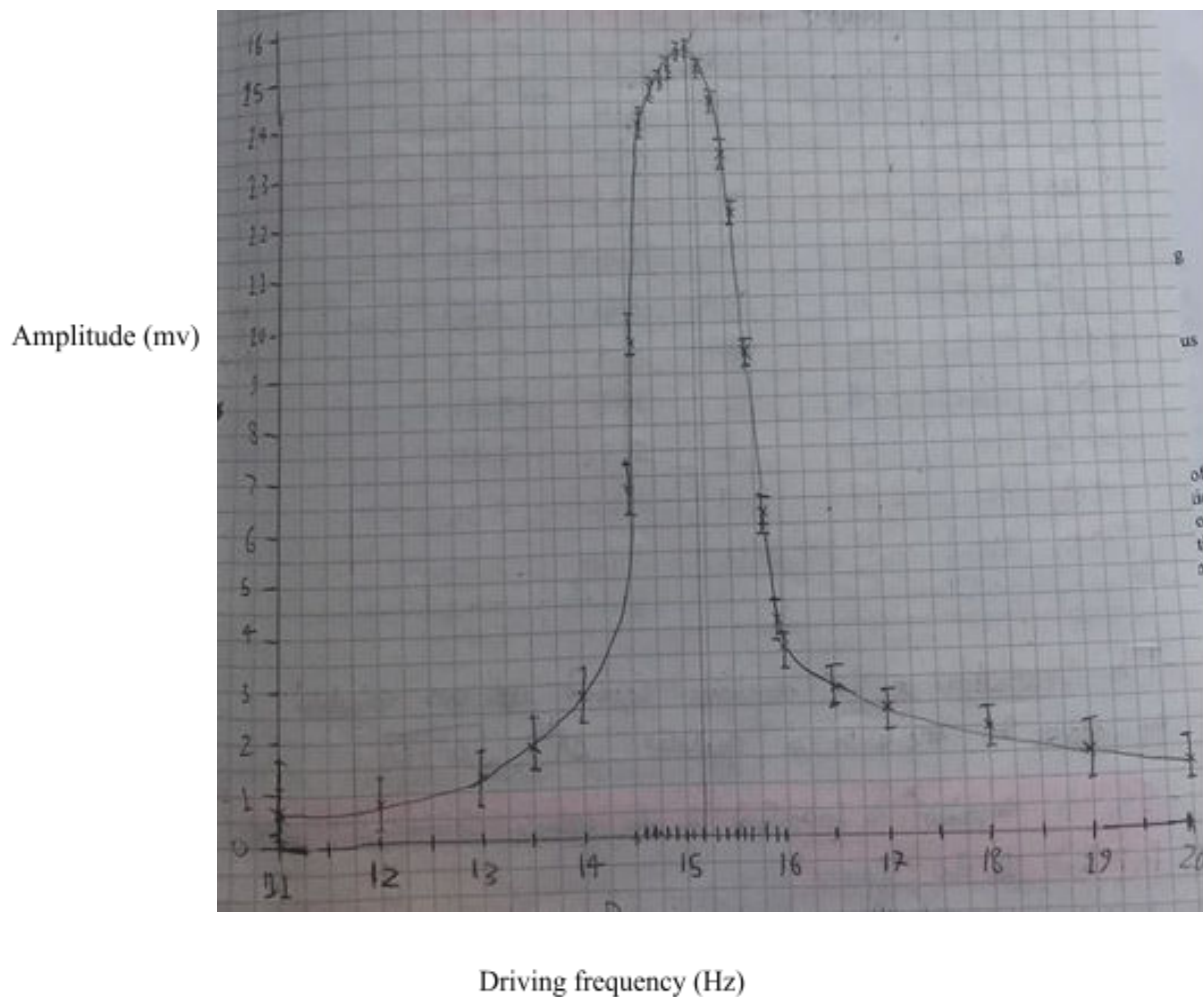


Figure 4.1. Graph of amplitude against driving frequency. The graph shows that as the driving frequency approaches the natural frequency of the bar the resonance occurs and the amplitude increases. The resonant frequency was found to be  $15.2 \pm 0.005 \text{ Hz}$ .

## 4.2 Power-resonance curve

When the system is in steady state, all the power being dissipated through the damping term is being replaced by the driving force. The damping term is proportional to the velocity,  $F = -bv$  and the power of system  $P = Fv$ , from this it can be seen that power is proportional to velocity squared and therefore also the square of the amplitude. The uncertainty in the square of the amplitude was calculated by adding the fractional uncertainty of the amplitude in quadrature.

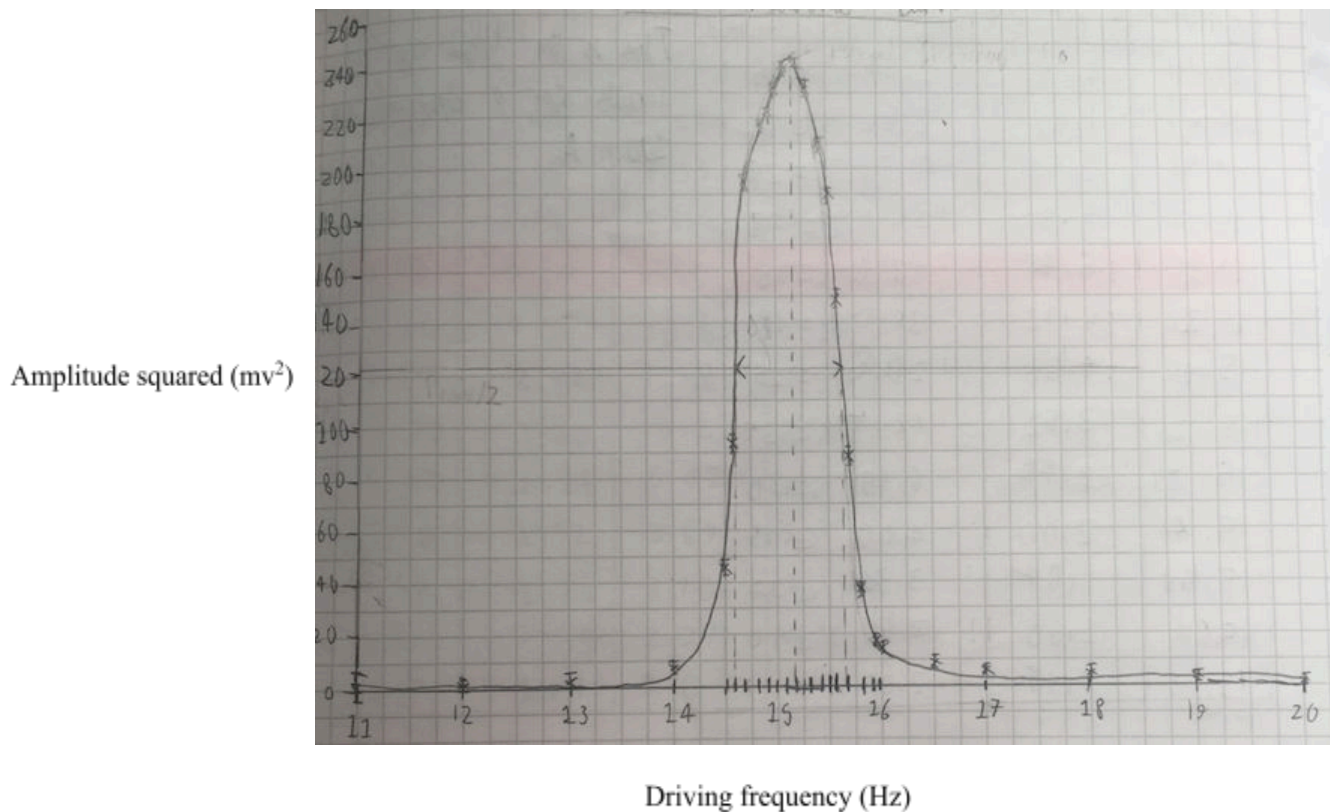


Figure 4.2. Graph of amplitude squared against driving frequency. The graph shows that as the driving frequency approaches the natural frequency of the bar the power of the system increases.

The natural frequency  $f^0$  was found to be  $15.2 \pm 0.005 \text{ Hz}$ , while the bandwidth  $\Delta f$  was found to be  $0.118 \pm 0.02 \text{ Hz}$ . The uncertainty in the bandwidth was calculated by the uncertainty in the ruler used to measure the width of the curve at half of its maximum. The quality factor was then calculated using Equation 2, with  $Q = 128.8 \pm 0.02$ .

The electromagnet was then energised with a current of 100mA. Figure 4.3 shows the graph for this level of damping.

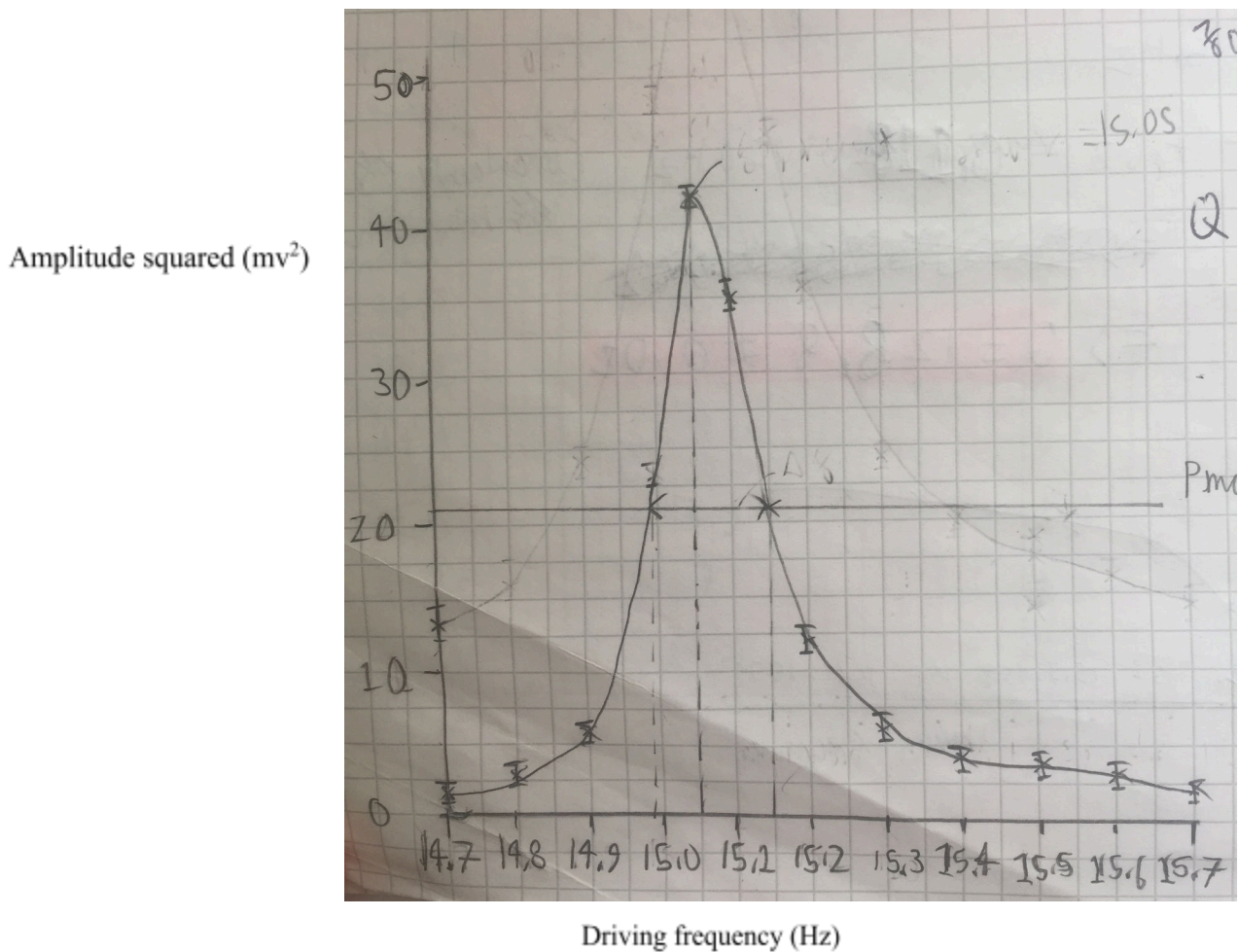


Figure 4.3. Graph of amplitude squared against driving frequency with damping of 100mA. The graph shows that damping results in a reduced maximum power at resonant frequency

The natural frequency for this level of damping  $f^0$  was found to be  $15.05 \pm 0.005\text{Hz}$ , while the bandwidth  $\Delta f$  was found to be  $0.155 \pm 0.02\text{Hz}$ . This increase in the bandwidth shows that the curve has flattened. The quality factor was then calculated using Equation 1, with  $Q = 97.1 \pm 0.02$ .



The damping force was increased by supplying the electromagnet with 200mA then 300mA of current. Figure 4.4 and 4.5 shows the graphs for this level of damping.

Amplitude squared ( $\text{mv}^2$ )

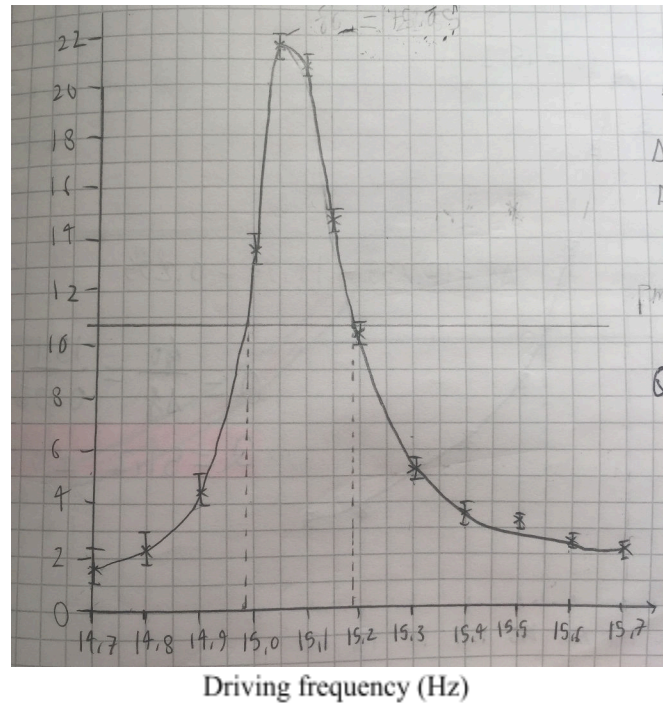


Figure 4.4. Graph of amplitude squared against driving frequency with damping of 200mA. The bandwidth was calculated to be  $0.208 \pm 0.02 \text{ Hz}$ , resulting in a quality factor of  $72.2 \pm 0.02$

Amplitude squared ( $\text{mv}^2$ )

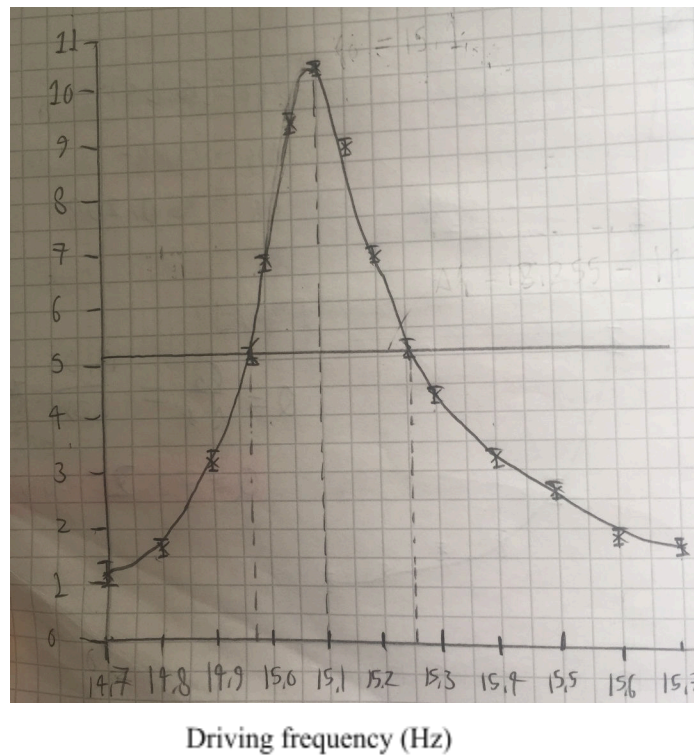


Figure 4.5. Graph of amplitude squared against driving frequency with damping of 300mA. The bandwidth was calculated to be  $0.294 \pm 0.02 \text{ Hz}$ , resulting in a quality factor of  $51.2 \pm 0.02$



### 4.3 Phase difference

The phase difference for between the driving frequency and the frequency of the oscillations was recorded using the two sine waves displayed on the oscilloscope. A plot is shown in figure 4.6.

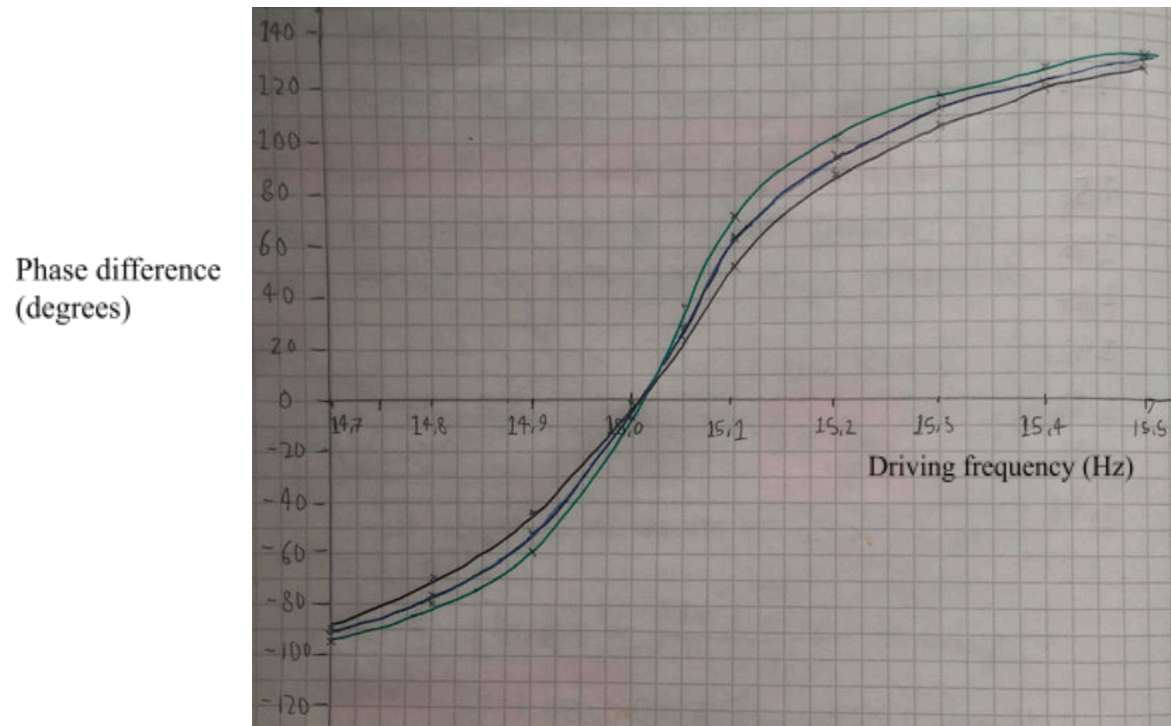


Figure 4.6. Graph of phase difference against driving frequency. The green line is for damping of 100mA, the blue line for damping of 200 mA, and the black for damping of 300mA.

Figure 4.6 shows that increased levels of damping decrease the rate at which the phase difference changes for a given change in the driving frequency. Damping decreases the response of the system to a change in the driving frequency.

#### 4.4 Free oscillations

Free oscillations means no driving force, so oscillations occur at the systems natural frequency. The bar was set in motion by tapping it, the amplitude of the oscillations decay exponentially due to light damping from the air. The natural log of the amplitude was taken according to Equation 3 and plotted in figure 4.7

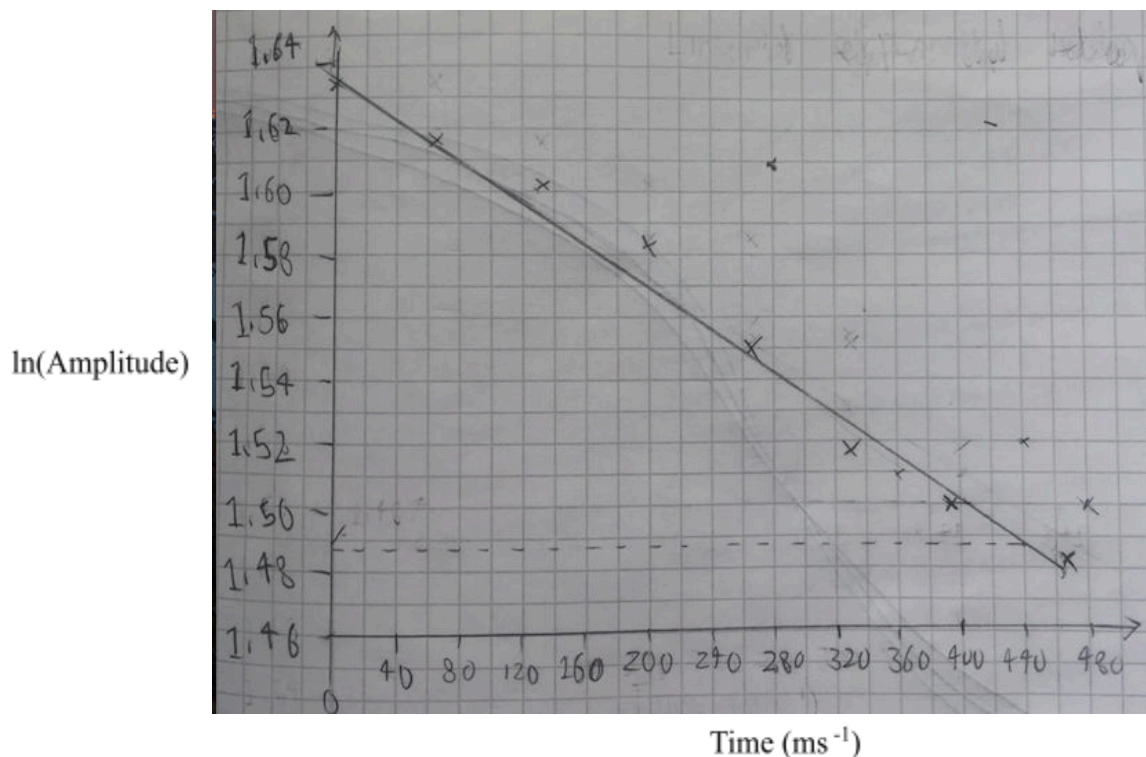


Figure 4.7. Linearised graph of exponential decay of oscillations. Natural log of amplitude against time. Gradient of graph gives  $-\gamma/2$  while y-intercept gives maximum amplitude

The gradient of the graph was calculated to be  $0.332 \pm 0.02$ , giving a value of  $\gamma = 0.664 \pm 0.02$ . The natural frequency of the bar,  $15.2 \pm 0.005 \text{ Hz}$  was converted into angular frequency and the quality factor was calculated using Equation 1, giving a value  $Q = 144 \pm 0.02$ .

#### 5. Conclusion

When the driving frequency approached the natural frequency of the bar, the amplitude increased by a factor of 30, displaying resonance. The quality factor of the undamped system was  $128.8 \pm 0.02$ , which fell to 97.1, 72.2 and 51.2 all with uncertainty  $\pm 0.02$  as the damping force on the system was increased. This fits with the theory of resonance as the quality factor is related to how fast energy is lost from the system, so increased damping should reduce it. The change in phase difference due to a change in the driving frequency was also reduced for a system with higher damping. The decay of free oscillation displayed an exponential decrease as expected as the gradient of the linearised equation was constant. A value was found for the decay constant  $\gamma = 0.664 \pm 0.02$ , the quality factor of the freely oscillating system was found to be  $144 \pm 0.02$ .

**References:**

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- [2] Israel D. Vagner, B.I. Lembrikov, Peter Rudolf Wyder, Electrodynamics of Magnetoactive Media, Springer, 2003, ISBN 3540436944
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